

# Life-cycle Management

At the end of this episode, I will be able to:

1. Explain typical life-cycle management components in modern networking

Leamer Objective: *Learn about common life-cycle management concepts as they apply to networking*

Description: In this episode, you will learn about typical life-cycle management processes and considerations that take place in modern networks.

- 
- Organizations will often use a framework when it comes to defining lifecycles for hardware and software. One example is the popular PPDIIO framework. This framework describes the lifecycle of networking technologies as follows:
    - Prepare - during this phase, there is time spent identifying requirements and outlining business goals
    - Plan - in this phase, the organization starts to actually plan the phases of the networking rollout
    - Design - here the details begin to take shape and things like budgets are started
    - Implement - here the networking technology begins to be deployed
    - Operate - once the previous phases have completed and there is a functional network technology that has been implemented, the operate phase can begin; here the technology is carefully monitored as it operates to ensure that resources are being used appropriately and business goals are being met
    - Optimize - in this final phase of the lifecycle; the organization strives to refine the design and operations to continue to bring superior business value to the organization
  - End-of-life (EOL) - EOL is a commonly used status for networking equipment refers to the point in time when a manufacturer discontinues the production of a specific product or product line. This decision is usually based on various factors such as technological advancements, market demand, and the need to allocate resources to newer products.
  - End-of-support (EOS) - EOS is a structured set of processes and policies that a manufacturer implements when discontinuing support for a specific product or product line. The primary goal is to help customers transition smoothly to newer solutions while providing a clear roadmap for the end of support lifecycle.
  - Software management - this discipline involves the phases of planning, development, testing, deployment, maintenance, and eventual retirement or replacement, ensuring systematic control and optimization of software applications throughout their existence.
    - Patches and bug fixes - patches and bug fixes are integral components of a software management lifecycle, addressing vulnerabilities, enhancing functionality, and resolving issues to ensure the ongoing stability, security, and performance of the software.
    - Operating system (OS) - Operating System (OS) updates are crucial elements of a software management strategy, delivering security enhancements, performance improvements, and new features to maintain the reliability and functionality of the system.
    - Firmware - firmware updates are essential software modifications specifically applied to the embedded code within hardware devices, enhancing functionality, fixing vulnerabilities, and ensuring optimal performance and compatibility.
  - Decommissioning - decommissioning involves the planned and systematic removal of outdated or no longer needed hardware or software, ensuring proper data migration or archival, and facilitating the transition to newer technologies or solutions, thereby maintaining the efficiency and security of the overall networking ecosystem. This process includes considerations for data integrity, regulatory compliance, and environmental impact to streamline the retirement of hardware or software components.

---

Additional Resources:

*A Guide to Network Lifecycle Management*

<https://www.techtarget.com/searchnetworking/tip/A-guide-to-network-lifecycle-management>